SUGARCREEK PACKING CO.: INSTALLING REVERSE OSMOSIS TECHNOLOGY TO ACHIEVE ENERGY, WATER, AND CHEMICAL COST SAVINGS FOR STEAM SYSTEMS

SOLUTION OVERVIEW

SugarCreek Packing Co. is a diversified and flexible food manufacturer that helps companies develop food solutions. A reverse osmosis (RO) membrane system was installed at SugarCreek's CCI facility to improve energy and water efficiency and reduce chemical treatment costs associated with operating the facility's steam system.

SECTOR	LOCATION	PROJECT SIZE	FINANCIAL OVERVIEW
Industrial	Cincinnati	418,000 Square Feet	$233,000

BACKGROUND

SugarCreek Packing Co. is a diversified and flexible food manufacturer helping some of the largest companies develop food solutions. The company operates seven manufacturing plants across the United States. The facility in Cambridge City, Indiana, (CCI) is approximately 418,000 square feet and produces 80 million pounds of product per year. A reverse osmosis (RO) membrane system was installed at SugarCreek's CCI facility to improve energy and water efficiency and reduce chemical treatment costs associated with operating the facility's steam system.

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The RO system removes mineral content before the make-up water goes into the feedwater tank, helping to keep the boiler running smoothly because of lower alkalinity. Comparing operational data from January 2021 and 2022 after the RO system was installed, the annual natural gas savings are estimated at 3.7% or 6,066 MMBtu and $18,000; and the annual chemical cost savings are estimated at 43% or $25,000.

SOLUTIONS

In addition to energy and cost savings, this project was driven by the need for increased steam generation at the CCI plant. The cost of the RO system was $233,000, and project implementation took eight months to complete. Energy savings from the RO system were achieved primarily through the reduction of boiler blowdowns from the use of higher-quality make-up water, which requires fewer chemicals to treat and prevents consumption of natural gas that would otherwise be used to heat new makeup water after boiler blowdowns. Before the RO system was installed, make-up water for the boiler system had a total dissolved solids (TDS) concentration of 700 ppm, and the blowdown cycle of concentration (COC) was around 4. The installation o the RO system reduced the concentration of TDS in the make-up water to 35 ppm. Thi theoretically increased the maximum COCs to 77, but for safety reasons, the COC is set at 55. Operation at higher COCs reduces blowdown, leading to less chemical loss and the amount of chemical required to prevent condensate corrosion was reduced due to the lower alkalinity of the water.

Cost Savings	Energy	Water	Chemical
2021 Annual Total	$495,811	$90,268	$68,552
2022 Annual Total (Actual and Normalized)	$477,466	$76,930	$42,679
2022 to 2021 Annual Savings	$18,345	$13,338	$25,873
2022 to 2021 Annual Savings (%)	3.7%	14.8%	37.74%

The project scope included both a steam system upgrade and an expansion of the building. SugarCreek's operations at CCI currently require 58,000 pounds per hour of steam to operate, but before this project, the system could generate only 38,000 pounds per hour. With the support of vendors and the SugarCreek onsite operation team, time was saved on testing and commissioning, and the system began normal operation in December 2021.

Figure 1: RO Skid

Figure 2: RO Concentration Tank

OTHER BENEFITS

The annual water savings from the CCI project is estimated at 11.1% or 3,340 kGal and $10,000. These savings are based on measurements taken during the month of January, so additional water savings will be realized during the summer months with RC water being used for cooling processes. SugarCreek is also planning projects that will offset water use with the concentrated water from the RO system for processes that are not sensitive to water hardness such as washing and rinsing.

Due to the success of the RO system at CCI, the SugarCreek engineering team is evaluating this technology for implementation at other locations. The plant located in Frontenac, Kansas, is another candidate for an RO system installation to improve overall boiler operation and steam production.